

SMART INDIA HACKATHON 2026 · NATIONAL SCREENING STAGE

DrishtiCare

Quality-aware, explainable diabetic-retinopathy screening for rural healthcare workflows
— built in MATLAB and Simulink.

PROBLEM STATEMENT

SIH 26038

SPONSOR

MathWorks

TEAM

6 members + mentors

SIH TEAM ID

Diabetic retinopathy is found late, because screening capacity is the constraint

WHERE THE LOAD ACTUALLY FALLS

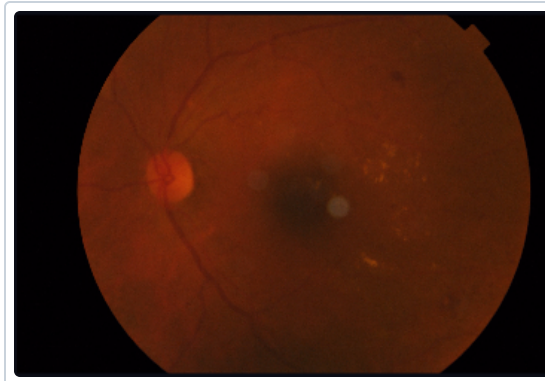
- At the planning assumption of 100,000 patients a year, our own measured operating point generates **153.6 specialist referrals per working day** — 38,406 a year.
- General ophthalmology is not the binding constraint. For vitreoretinal grading it is retina specialists, at roughly **one per 1.26 million** population.
- So the useful question is not "can a CNN grade a retina". It is **what does the system do with an image it should not answer?**

OUR ANSWER

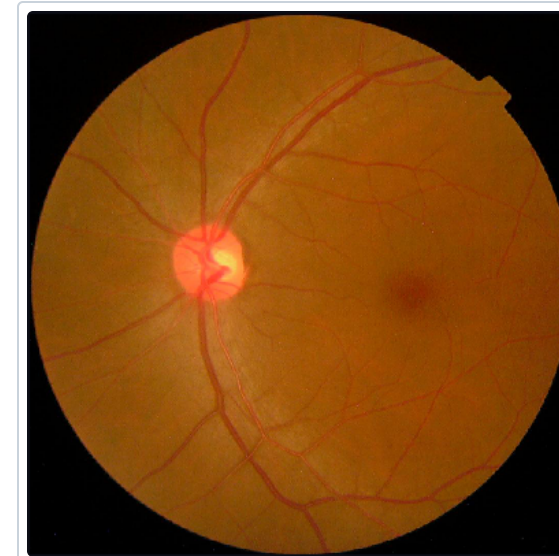
A screening workflow whose **quality gate runs before the model**. A poor or unfamiliar image is withheld for recapture rather than graded. That refusal behaviour — not the classifier — is the product.

Referral demand: data/analysis/simulink_resource_simulation/scenario_results.csv, row D100.

Workforce constraint: Vashist et al. 2021, citing WHO 2020. Full citations on the impact slide.



Withheld. Quality FAIL. The classifier is never invoked, no grade is produced, and the case is routed to recapture or manual review.

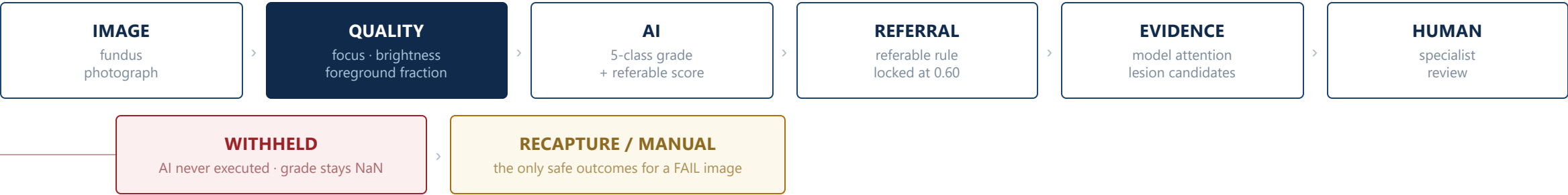


Graded. Quality PASS. The image proceeds to grading, referral routing and evidence.

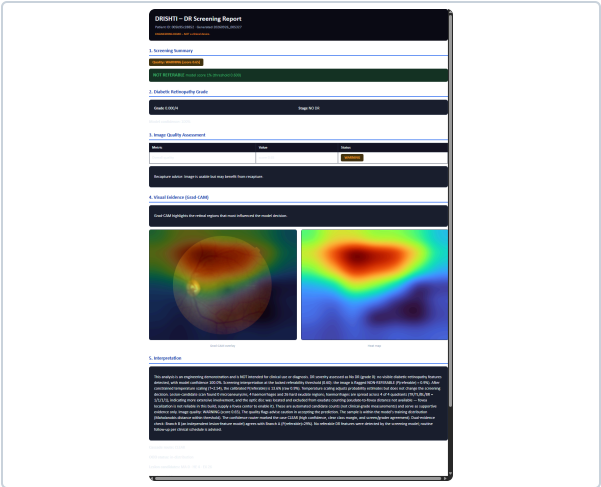
Both images are real fundus photographs from the locked APTOS validation split, captured by the running application (results/visual_qa/).

The quality gate runs before the AI, not after it

This ordering is the whole design. A screening system that grades every image it is handed has no way to say “I don’t know”.



- **Quality FAIL is a hard gate, not a display setting.** The classifier is never invoked, no Grad-CAM is computed, and the output carries an explicit withheld state rather than a silent blank.
- **Referral is a locked rule.** Referable is $P(\text{referable}) \geq 0.60$. The threshold is provenance-pinned and was not retuned.
- **Uncertainty is routed, not averaged away.** The cascade router sends borderline cases to REVIEW and the least confident to ABSTAIN.



The branded A4 report is a real artifact written to disk from the same result struct that drives the screen — seven sections including the non-clinical disclaimer.

What is built and measured, and what is not

BUILT	Running, measured, verified	Hardening programme P0–P25: 230/230 checks PASS	PLANNED / BLOCKED	Named blocker, named owner	Not hidden, not silently dropped
	Image quality gate — focus, brightness, foreground fraction. 3,662 images measured: PASS 65.48% / WARNING 26.68% / FAIL 7.84%.			External validation — BLOCKED, not performed. Messidor-2 needs an ADCIS registration; Sin-NP DR 2019 needs a data request. Both require a human download we cannot perform.	
	5-class DR grader — ResNet-18, EyePACS-pretrained, class-balanced. QWK 0.8914 on the locked 733-image split.			Owner: team, on dataset licensing. The harness is written and passes 6/6; on the locked split it reproduces sensitivity 0.9060 / specificity 0.9471 / ROC-AUC 0.9796 / PR-AUC 0.7821 exactly, so the metric path is faithful.	
	Binary referable classifier — sensitivity 0.9060 / specificity 0.9471 at the locked 0.60 threshold.			Minority-recall experiments — deliberately switched off. Target-boosted weighting and targeted augmentation on Severe and Proliferative are disabled in the training script by operator decision, not overlooked.	
	OOD advisory detector — Mahalanobis, locked at its p99 of 34.22. Surfaces in the narrative; never alters a route.			Owner: mentor approval.	
	Cascade router — CLEAR 648 / REVIEW 75 / ABSTAIN 10 on the validation split.			Grad-CAM localisation agreement — measured, and weak. Saliency-in-lesion 3.1%, pointing game 7.4%, mean IoU 0.035. MACE was not measured.	
	Temperature calibration — T = 2.5382, display-only.			We did not retrain to chase a better number. Documented as a limitation.	
	Grad-CAM, lesion evidence, Branch B second opinion — running, with their measured weaknesses documented.			Fovea localisation — failed. 0 of 10 held-out images within 300 px. The pipeline therefore emits no fovea-derived distance.	
	Dashboard and branded A4 report — headless dashboard verification 35/35 PASS; report verification PASS.			Revisit if fovea-labelled data appears.	
	Simulink district model — a genuine .slx reproducing its reference engine exactly over 250 days; 11/11 sanity checks PASS.			Image enhancement — excluded on evidence. The A/B experiment showed it degraded grading, so the deployed path feeds the model the raw resized image and shows enhancement as a display aid only.	
	Unseen-input rehearsal harness — ran end to end on 19 foreign-camera images, 4/4 contract checks PASS.			Decision recorded, not revisited.	
				Vessel segmentation in production — excluded on evidence. Adding vessel features reduced Branch-B AUC from 0.8969 to 0.8810, so the branch ships without them.	
				Negative result kept in the record.	

MEASURED PERFORMANCE

Locked 733-image validation split, seed 42

Every figure below is frozen under a written contract: no retuning of the threshold or the calibration temperature, and no use of the sealed official test set.

0.8281
Accuracy

0.6805
Macro F1

0.8914
Quadratic weighted kappa

0.9060
Sensitivity, referable DR

0.9471
Specificity, referable DR

0.9796
ROC-AUC

0.7821
PR-AUC

0.60
Referral threshold — locked,
provenance-pinned

2.5382
Calibration temperature — display-only

733
Validation images (2,929 train, seed 42)

PER-CLASS RECALL — WHERE THE WEAKNESS IS

No DR	0.9834	
Mild NPDR	0.6081	
Moderate NPDR	0.7850	
Severe NPDR	0.4872	weak — main 5-class weakness
Proliferative DR	0.5254	weak — experiments deliberately off

OPERATIONAL BEHAVIOUR, MEASURED

QUALITY
GATE, N
= 3,662
65.48 /
26.68 /
7.84 %

CLEAR /
REVIEW
/
ABSTAIN
648 / 75
/ 10

OOD
BOUND
(LOCKED
P99)
34.22

MODEL-
ONLY
INFERENCE
0.1028
s/img

FULL
DEMO
PATH,
MEDIAN
11.03
s/img

VERDICT ON
APTOS
VALIDATION
internal,
single-
dataset

The two latency figures are deliberately not conflated: 0.1028 s/img is model screening only, 11.03 s/img is the complete demo path. High-resolution fundus photographs are slower again — a judge supplying a full-resolution image should expect roughly 30–45 s.

It refuses. We tested that on images it had never seen.

Every automated check in this project had run on the same split the champions were selected on. We then ran the full pipeline end to end on **19 unseen fundus images from three foreign cameras** — DRIVE test, the IDRiD B testing set, and real-world web fundus from DRIMDB. Nothing was retrained or tuned, the 0.60 rule was untouched, and the model hashes are unchanged.

19 / 19

completed. No crash, no unhandled exception.

4 / 4

contract checks PASS: valid route band, quality FAIL never auto-answered, locked 0.60 rule still holds.

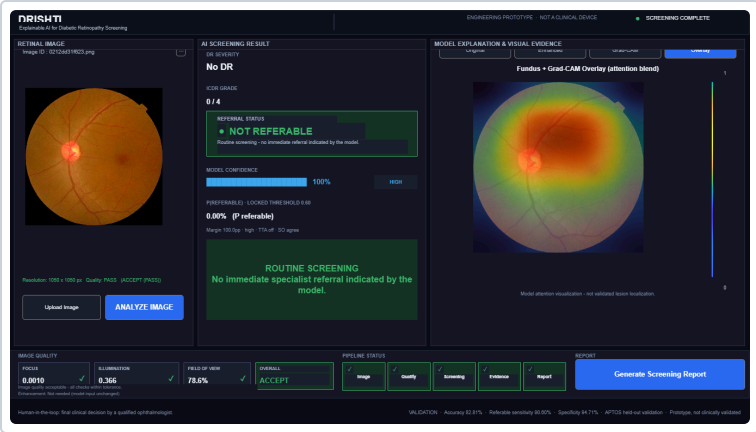
8

quality FAIL images — every one withheld *before* any model execution.

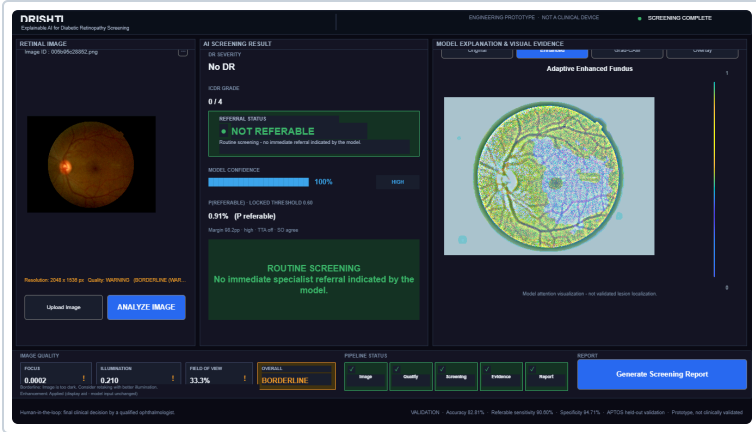
42%

of unseen images were withheld, against 7.84% FAIL on APTOS. The gate over-rejects foreign cameras.

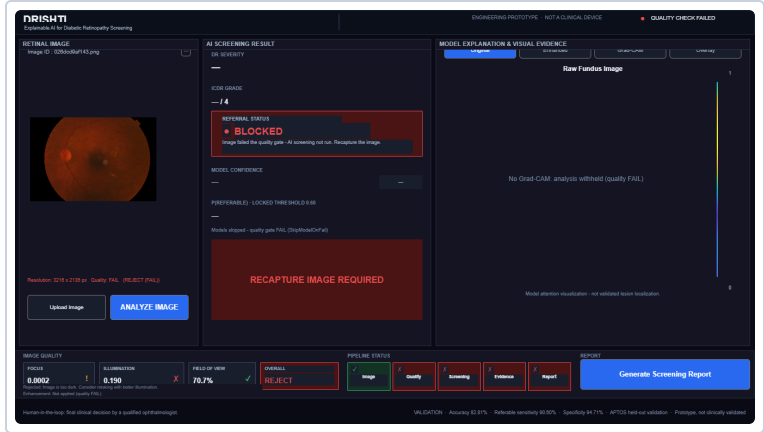
- **The safety contract is not an APTOS artefact.** Every quality FAIL was withheld before the model ran, and the locked decision rule was reproduced on every image that did reach the classifier.
 - **The OOD detector genuinely fires out of distribution.** It flagged unseen images above its locked 34.22 bound. It remains advisory only and altered no route in this run.
 - **Over-refusal is the safe direction to fail** — a withheld image is never auto-answered. Expect “AI grading skipped, please recapture”: that is the designed behaviour.
- The honest negative finding.** One DRIMDB image the dataset itself labels as ungradable passed the quality gate and came back a confident referable grade, routed CLEAR, with its OOD distance just under the bound. Neither gate caught it. We are not tuning it away — the threshold and the OOD bound are provenance-pinned. We state it, and we demo accordingly. On out-of-distribution real-world imagery, this system is not perfect.



PASS — graded and routed, with confidence and evidence.



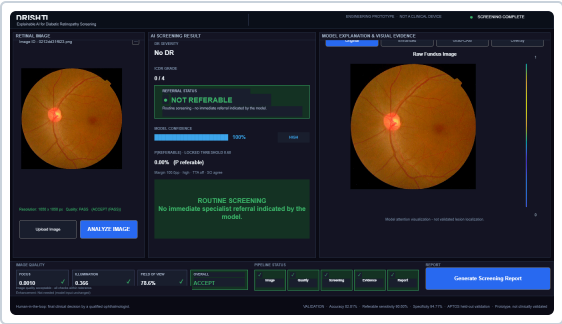
WARNING — proceeds, but flagged as borderline.



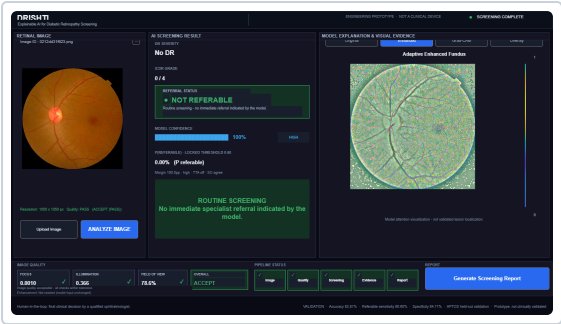
FAIL — AI grading skipped; the model is never called.

Grad-CAM shows model attention. It is not lesion localisation.

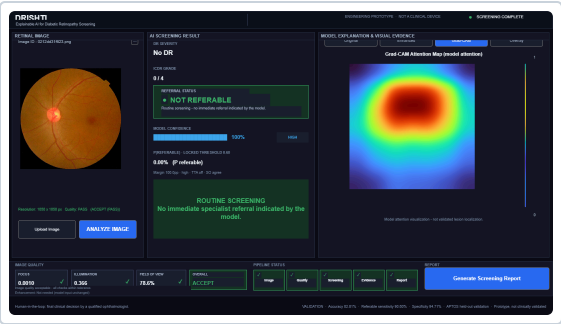
Four views of one real case from the running application. The heat-map is a statement about *where the model looked*, not about *where a lesion is* — and we measured how far apart those two things are.



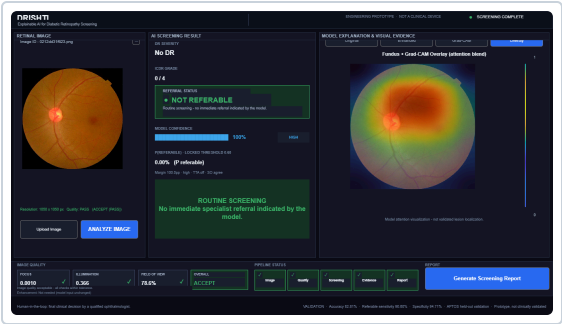
Original — the model input, unchanged.



Enhanced — display aid only. Not the model input.



Grad-CAM — regions that most influenced the decision.



Overlay — for the reviewer's eye, not a finding.

WE MEASURED OUR OWN EXPLAINABILITY AGAINST IDRID LESION MASKS

3.1%

saliency mass inside a lesion mask

7.4%

pointing-game accuracy

0.035

mean IoU, top-20% pixels

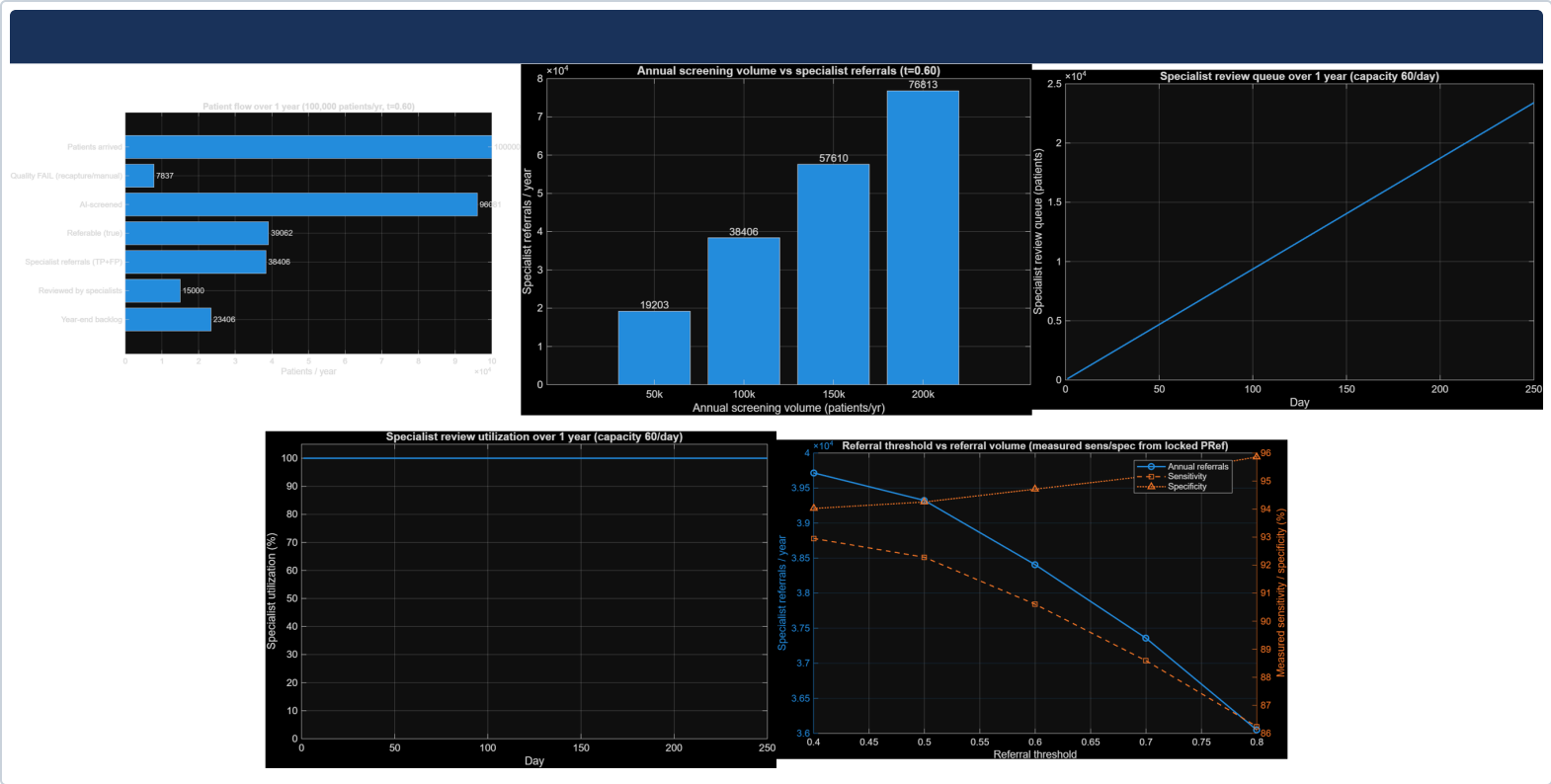
n/m

MACE — not measured

Against an areal chance baseline of 0.0220, that is about 1.39× chance. The alignment is weak, and we did not retrain the model to improve a visualisation metric. We publish the number because a screening prototype that overstates its own evidence is not deployable.

- **Say “model attention”, not “lesion localisation”.** The second claim is not supported by our own measurement and we do not make it.
- **Lesion counts are candidates, not findings.** The microaneurysm detector has patch AUC 0.976 but detection recall of only 0.113, and the classical detectors over-fire on unfamiliar images.
- **Calibration is presentation-only.** T = 2.5382 makes the displayed confidence usable; it is not evidence of external generalisation.

The bottleneck the model exposes is specialist review, not AI processing



Five panels from one simulation: patient flow, referral demand, queue growth over the year, utilisation against the capacity scenarios, and workload by referral threshold.

Engineering resource-planning simulation. Not clinical validation and not a staffing prescription. The measured input is our locked-split operating point (sensitivity 0.9060, specificity 0.9471 at threshold 0.60). Patient volume, prevalence, specialist capacity and the recapture rate are labelled assumptions, not measurements. The .s1x reproduces its reference engine exactly over 250 days; 11/11 sanity checks pass.

AT 100,000 PATIENTS PER YEAR

IMAGES REACHING AI SCREENING

96,081

REFERRALS GENERATED

38,406 39.97% of screened

NEVER CONSUME SPECIALIST TIME

60.03%

REFERRALS PER WORKING DAY

153.6

SPECIALISTS IMPLIED, 20 EXAMS/REVIEWER/DAY

7.68

THE BREAK POINT

At 60 reviews/day the year ends at a 23,406 backlog. At 120/day, still 8,406. Only at **~180 reviews per day** does the queue clear, at 85.3% utilisation. A second, independent discrete-event simulation reaches the same conclusion.

Three national statistics, each carrying its source and its year

We verified every figure we quote. Where our own earlier draft did not survive verification, we say so on the slide rather than quietly swapping it.

74.2 million

Indian adults aged 20–79 with diabetes, 2021. India's own ICMR-INDIAB-17 national survey puts the same year nearer **101 million** — the two use different methods and are not interchangeable, so we name the body.

IDF Diabetes Atlas, 10th ed. 2021 — Sun H. et al., *Diabetes Res Clin Pract* 2021;183:109119, doi:10.1016/j.diabres.2021.109119. ICMR-INDIAB-17 — Anjana R.M. et al., *Lancet Diabetes Endocrinol* 2023, doi:10.1016/S2213-8587(23)00119-5.

SUPERSEDED: THE WIDELY QUOTED 77 MILLION FIGURE IS THE IDF 2019 ESTIMATE (9TH ED.) AND HAS BEEN REVISED TWICE SINCE. DO NOT PRESENT IT AS CURRENT.

16.9% – 18.1%

Roughly **1 in 6 to 1 in 5 Indians with diabetes aged 50 and over** have diabetic retinopathy. We give the range because two national-survey-grade sources disagree, and the honest reading is a band rather than a decimal.

18.1% (95% CI 14.8–21.4), age ≥50 — Jotheeswaran A.T. et al., *Indian J Endocrinol Metab* 2016;20(Suppl 1):S51–S58, doi:10.4103/2230-8210.179774. 16.9% (95% CI 15.9–17.9) — MoHFW/NPCB National Survey 2015–19, Vashist P. et al., *Indian J Ophthalmol* 2021, doi:10.4103/ijo.IJO_1310_21.

NOT DEFENSIBLE FOR ALL ADULTS: THE SAME META-ANALYSIS GIVES 14.9% FOR AGE ≥30, AND POPULATION PREVALENCE IS FAR SMALLER AGAIN. THE AUTHORS THEMSELVES FLAG THEIR OWN DENOMINATOR AS WEAK.

1 : 65,221

Indian ophthalmologists per person, about 15 per million, nationally. The binding constraint for automated *vitreoretinal* grading is roughly **19× worse**: about one **retina specialist per 1.26 million** people.

Vashist P. et al., “Human resources and infrastructure for ophthalmic services in India”, *Indian J Ophthalmol* 2025;73(11):1679–1686, doi:10.4103/IJO.IJO_2816_24; data collected 2020–21, 20,944 ophthalmologists. Retina specialist density: WHO 2020, cited in Vashist et al. 2021.

VERIFICATION NOTE: A LOWER RURAL SCARCITY FIGURE IN OUR OWN EARLIER DRAFT DID NOT SURVIVE CHECKING — IT TRACES TO A VISION 2020 PLANNING TARGET FOR ASIA, NOT AN INDIAN COUNT. WE USE THE MEASURED NATIONAL RATIO INSTEAD.

LOCKED VALIDATION MEASURED 733 held-out images	QUALITY AUDIT MEASURED 3,662 images	UNSEEN-INPUT REHEARSAL 19 images, 3 cameras	DISTRICT SIMULATION 250-day horizon	HARDENING PROGRAMME 230/230 checks PASS
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What this system cannot do yet

These are on the slide, not in the appendix. A prototype that hides its measured negatives is not safe to hand to a screening programme.

ITEM	WHAT WE MEASURED	STATUS	BLOCKER OR OWNER
External validation	Not performed. The reported 0.9060 / 0.9471 / 0.9796 / 0.7821 are APTOS-internal, on a single dataset.	BLOCKED	Messidor-2 (ADCIS registration) and Sin-NP DR 2019 (data request) both need a human download. Harness written, 6/6 checks pass.
Grad-CAM vs lesion masks	Saliency-in-lesion 3.1%, pointing game 7.4%, mean IoU 0.035, about 1.39× areal chance.	MEASURED WEAK	MACE not measured. We did not retrain to improve a visualisation metric.
Confident grade on an ungradable image	One DRIMDB image the dataset labels Bad passed the quality gate and came back referable, routed CLEAR. Neither gate caught it.	OPEN	Recorded, not tuned away — the threshold and OOD bound are provenance-pinned.
Fovea localisation	0 of 10 held-out images within 300 px of ground truth.	FAILED	No fovea-derived distance is emitted; the field stays unavailable with an explicit narrative line.
Severe / Proliferative recall	0.4872 and 0.5254 — the main 5-class weakness.	OFF BY DECISION	Target-boosted weighting and targeted augmentation are disabled in the training script. Owner: mentor approval.
Microaneurysm detection	Patch AUC 0.976, but detection recall only 0.113 at the evaluated threshold.	LOW RECALL	Counts are reported as candidates, never as findings.
Optic-disc generalisation	Locates 9/10 within 300 px on IDRiD, but only 10.6% acceptance on APTOS validation.	LIMITED	Trained on a different camera set; the honest hardware-threshold refusal works as designed.
Image enhancement	The A/B experiment showed it degraded grading.	EXCLUDED ON EVIDENCE	Excluded from the inference path; retained as a display aid only.
Vessel segmentation	Adding vessel features moved Branch-B AUC from 0.8969 to 0.8810.	EXCLUDED ON EVIDENCE	Measured negative result, kept in the record rather than forced into production.

Everything here is reproducible from a public repository

ENTRY

- **Problem statement** SIH 26038 · sponsor MathWorks
- **Team** 6 members + mentors · SIH Team ID

- **Repository** github.com/Aryan41211/DrishtiCare
- **Stack** MATLAB and Simulink; ResNet-18 transfer learning, EyePACS-pretrained, class-balanced
- **Launch** `launchRetinaAI.m` from the repository root
- **Governance** frozen model contract with SHA-256 hashes, a decision log, and a per-task evidence record

DATASETS, STATED PLAINLY

- **APTOS 2019** — grading data. Used.
- **IDRiD** — lesion and optic-disc ground truth. Used for evaluation, not for classifier selection.
- **DRIVE** — vessel study. Used.
- **DRIMDB** — downloaded and licence-cleared, used *only* in the unseen-input rehearsal. Not used in any reported result.
- **Messidor-2, Sin-NP DR 2019** — **not obtained**.

DrishtiCare is an engineering prototype for Smart India Hackathon 2026. Not a clinical device. No clinical validation has been performed. Not for diagnosis, treatment or triage decisions. External validation remains blocked.

REFERENCES

Vashist P. et al. (2021)	National survey of DR, 21 districts, n = 5,986 examined. <i>Indian J Ophthalmol</i> 69(11):3087–3094. doi:10.4103/ijo.IJO_1310_21
Jotheeswaran A.T. et al. (2016)	DR prevalence by age band, systematic review. <i>Indian J Endocrinol Metab</i> 20(Suppl 1):S51–S58. doi:10.4103/2230-8210.179774
Vashist P. et al. (2025)	Ophthalmic workforce and infrastructure, national survey. <i>Indian J Ophthalmol</i> 73(11):1679–1686. doi:10.4103/IJO.IJO_2816_24
Sun H. et al. (2021)	IDF Diabetes Atlas 10th edition, India estimate. <i>Diabetes Res Clin Pract</i> 183:109119. doi:10.1016/j.diabres.2021.109119
Anjana R.M. et al. (2023)	ICMR-INDIAB-17 national diabetes survey. <i>Lancet Diabetes Endocrinol</i> . doi:10.1016/S2213-8587(23)00119-5
Pradeepa R., Mohan V. (2021)	Tabulation of the IDF 2019 India estimate. <i>Indian J Ophthalmol</i> 69(11):2932–2938. doi:10.4103/ijo.IJO_1627_21

How to read every number in this deck. Each is traceable to a committed artifact — the frozen contract, the validation reports, the rehearsal record, the simulation CSV, or a cited publication with a DOI. Where a figure was only provisional, we labelled it provisional rather than dropping the caveat. Where our own earlier draft did not survive verification, we left it visible and marked it unsupported. We would rather show a weaker number we can defend than a stronger one we cannot.